# diff - diff

**Callaway-Sant'Anna. Cengiz-style stacking.**

**BJS-style imputation. The classic 2x2.**

# One regression to rule them all.

**LP-DiD: Local Projections DiD.**

*Now in diff-diff.*

*Dube, Girardi, Jordà & Taylor (2025).*

Journal of Applied Econometrics 40(5)

**diff-diff v3.6.1**

# The staggered revolution

# left you with a shelf.

### Classic 2x2 DiD

Where you started.

One clean comparison.

### Callaway-Sant'Anna

Group-time ATTs.

Its own aggregation layer.

### Stacked DiD (Cengiz et al.)

Dataset duplication.

Its own weighting scheme.

### BJS-Style Imputation

Borusyak-Jaravel-Spiess.

Its own two-step machinery.

*Three fixes for TWFE's negative weights - plus the textbook case they outgrew.*

**Same causal question. Four mental models.**

# Same question.

# Four different machines.

### Staggered State Policy

Minimum wage rolls out state by state. CS? Stacked? Which aggregation?

### Phased Product Rollout

The feature ships to markets in waves. Event study? Which estimator?

### Programs That Switch Off

The retention treatment ends. Absorbing-only estimators don't apply.

### Survey-Weighted Panels

CPS-style pweights, strata, PSUs. Which estimator takes a design?

# LP-DiD.

*It's just a regression.*

A local projection per horizon: long difference,

time fixed effects, clean controls. Plain OLS.

**One long difference per horizon.**

OLS on clean controls. You can read every comparison it makes.

**Weights provably non-negative.**

Clean-control baseline result (Eqs. 9-10). With covariates, prefer RA.

**The estimand is a dial.**

Variance-weighted for precision, or equal-weighted - your call.

*Local projections (Jordà 2005), pointed at DiD.*

# Clean controls,

# horizon by horizon.


**Each horizon is one regression you could run by hand.**

*In the absorbing (default) path, already-treated units never serve as controls.*

# The zoo was one regression

# all along.

$$y_{i,\,t+h} - y_{i,\,t-1} \;=\; \beta_h\,\Delta D_{it} \;+\; \delta_t^{\,h} \;+\; e_{it}^{\,h}$$

$$E(\beta_h) \;=\; \sum_{g \neq 0} \omega_{g,\,h}\,\tau_h^{g}\,, \qquad \omega_{g,\,h} \geq 0$$

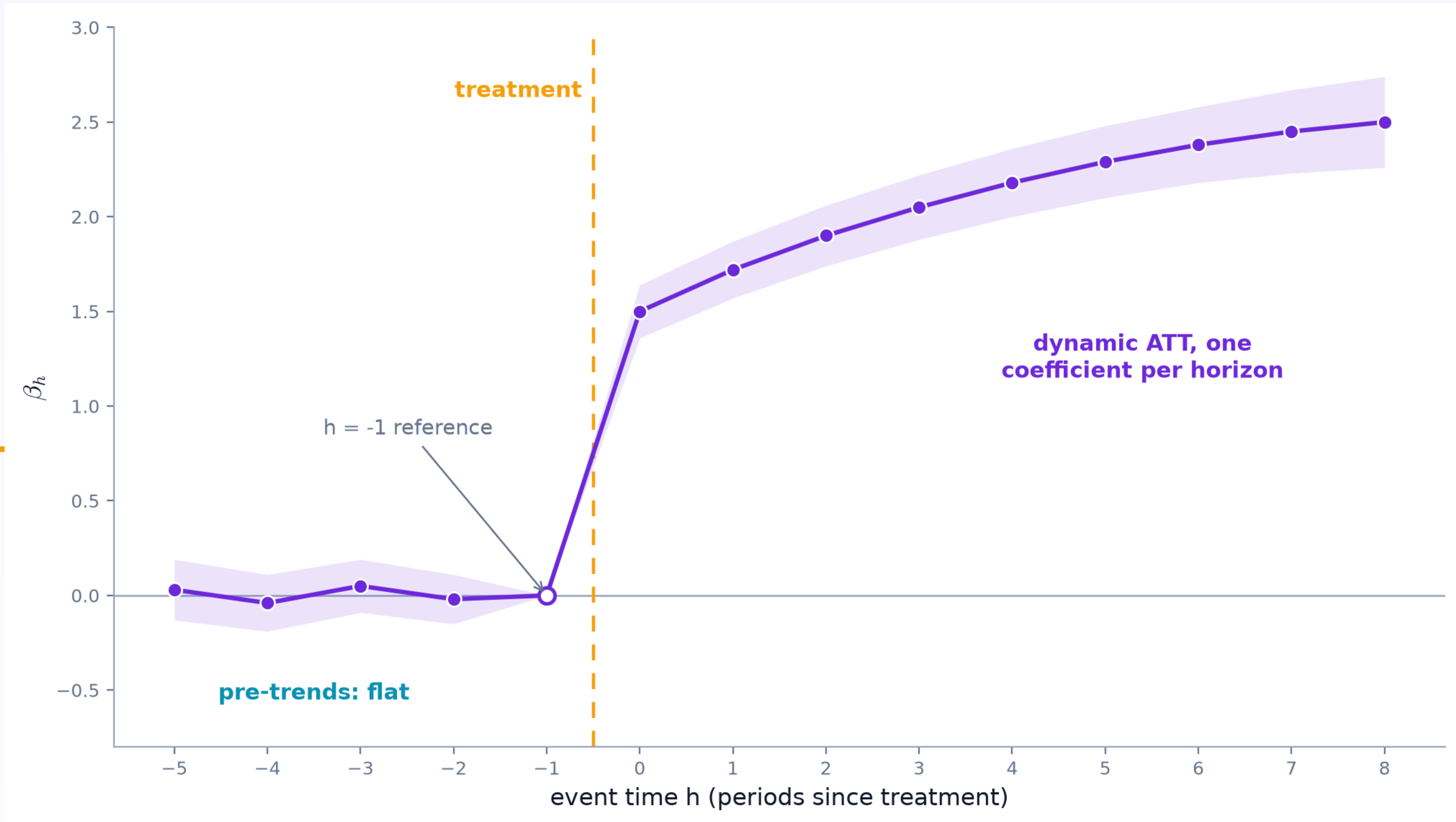
always non-negative -- no forbidden comparisons

| | | |
|---|---|---|
| `reweight=True` | == **Callaway-Sant'Anna** | *exact (Sec. 3.7)* |
| `default (variance-weighted)` | == **Cengiz et al. stacked** | *exact (Sec. 3.7)* |
| `h=0, single cohort` | == **classic 2x2 DiD** | *exact (Sec. 2.2)* |
| `pmd='max'` | -> **BJS imputation** | *exact single-cohort (fn. 10-11)* |

*Numerical equivalences proven in the paper - not analogies.*

# Pre-trends are just

# negative horizons.




**The event study is native - not a post-hoc aggregation.**

# The Code.

Same sklearn-like API as every diff-diff estimator.

```
from diff_diff import LPDiD

result = LPDiD(
    pre_window=5, post_window=10,
).fit(
    data,
    outcome='revenue', unit='store',
    time='week', treatment='treated',
)

print(result.att)        # pooled post ATT = 2.31
result.event_study       # one beta per horizon

# flip one switch:
#   reweight=True   ->  Callaway-Sant'Anna
#   non_absorbing=  ->  on/off treatments (entry effects)
#   survey_design=  ->  pweights + strata + PSUs (default path)
```

Callaway-Sant'Anna is literally a keyword argument.

# Production-ready.

### Cluster-Robust SEs

Unit-level by default,

matches Stata lpdid

### Covariates, Two Ways

RA path (BJS-style) or

direct with guardrails

### PMD Baselines

Premean differencing +

pooled pre/post ATTs

### Non-Absorbing Treatment

First-entry & effect-

stabilization (Eq. 12/13)

### Survey Designs

pweights, strata, PSU, FPC;

Binder TSL (default path)

### Composition Control

no_composition fixes the

post-window sample

*Composable where validated - survey_design runs the default path only.*

# Validated.

*Against the authors' own tooling - and against our own shelf.*

**Authors' R recipes (danielegirardi/lpdid)**

Event-study and pooled estimands match to ~1e-12.

**Independent fixest reconstruction**

Non-absorbing Eq. 12/13, variance-weighted: point + SE to ~1e-13.

**survey::svyglm, end to end**

Default-path survey: per-horizon point, SE, and df all pinned.

**Equivalences tested, not cited**

reweight == our CallawaySantAnna; PMD == our ImputationDiD (1 cohort).

*Now in diff-diff.*

# LP-DiD.

```
$ pip install --upgrade diff-diff
```

**github.com/igerber/diff-diff**

**diff - diff**

Difference-in-Differences for Python